

Deployment of PrestaShop on AWS EC2 Instance

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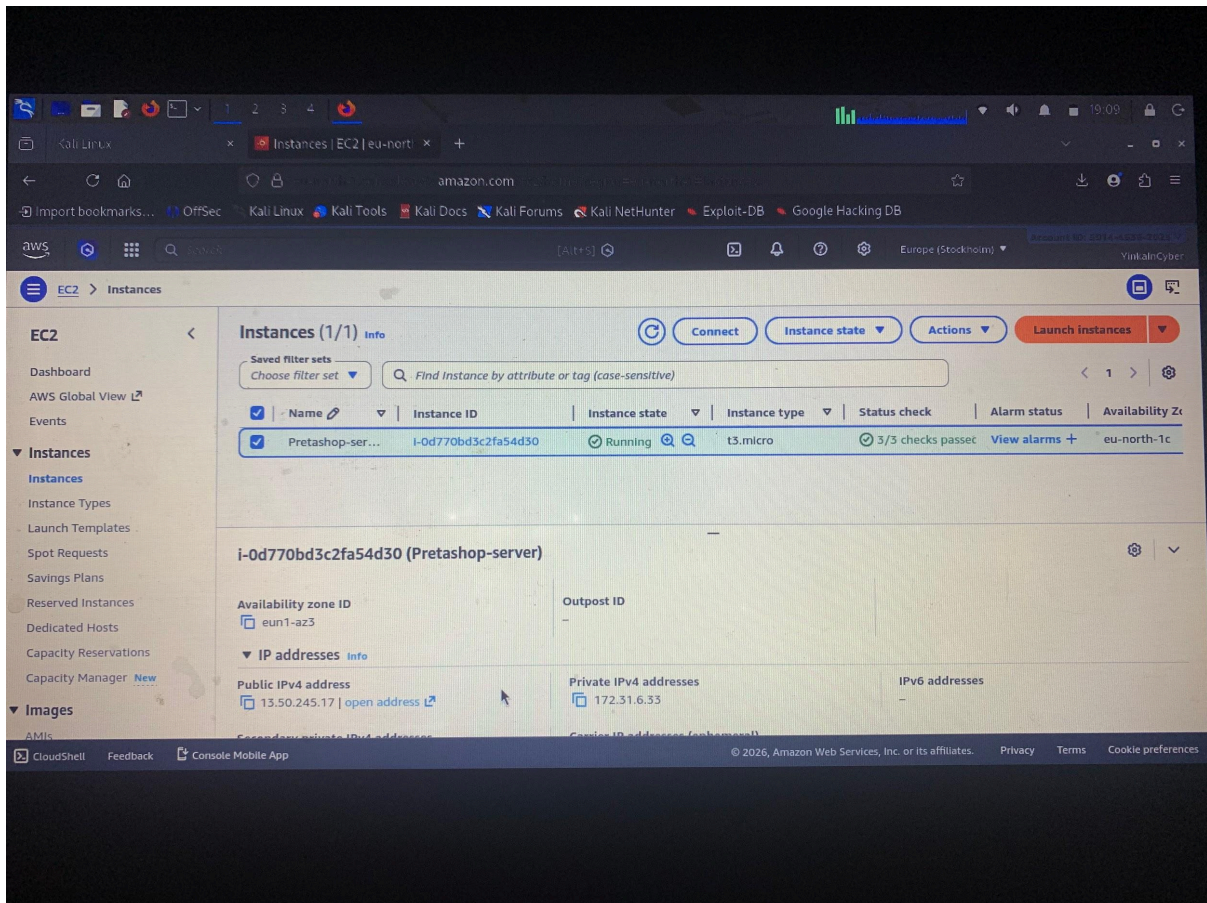
1. Introduction

This project demonstrates the deployment of an open-source e-commerce application, PrestaShop, on an AWS cloud infrastructure using an EC2 Ubuntu instance. The objective is to create a publicly accessible web server using AWS Free Tier services while applying basic cloud security and networking configurations.

2. AWS EC2 Instance Setup

An EC2 instance was launched using the AWS Free Tier with the following configuration:

- Instance Name: Prestashop-server
- Operating System: Ubuntu Server (Latest LTS)
- Instance Type: t2.micro (Free Tier eligible)
- Key Pair: SSH key pair generated during instance creation
- Public IP Address: Assigned automatically by AWS



3. Security Group Configuration

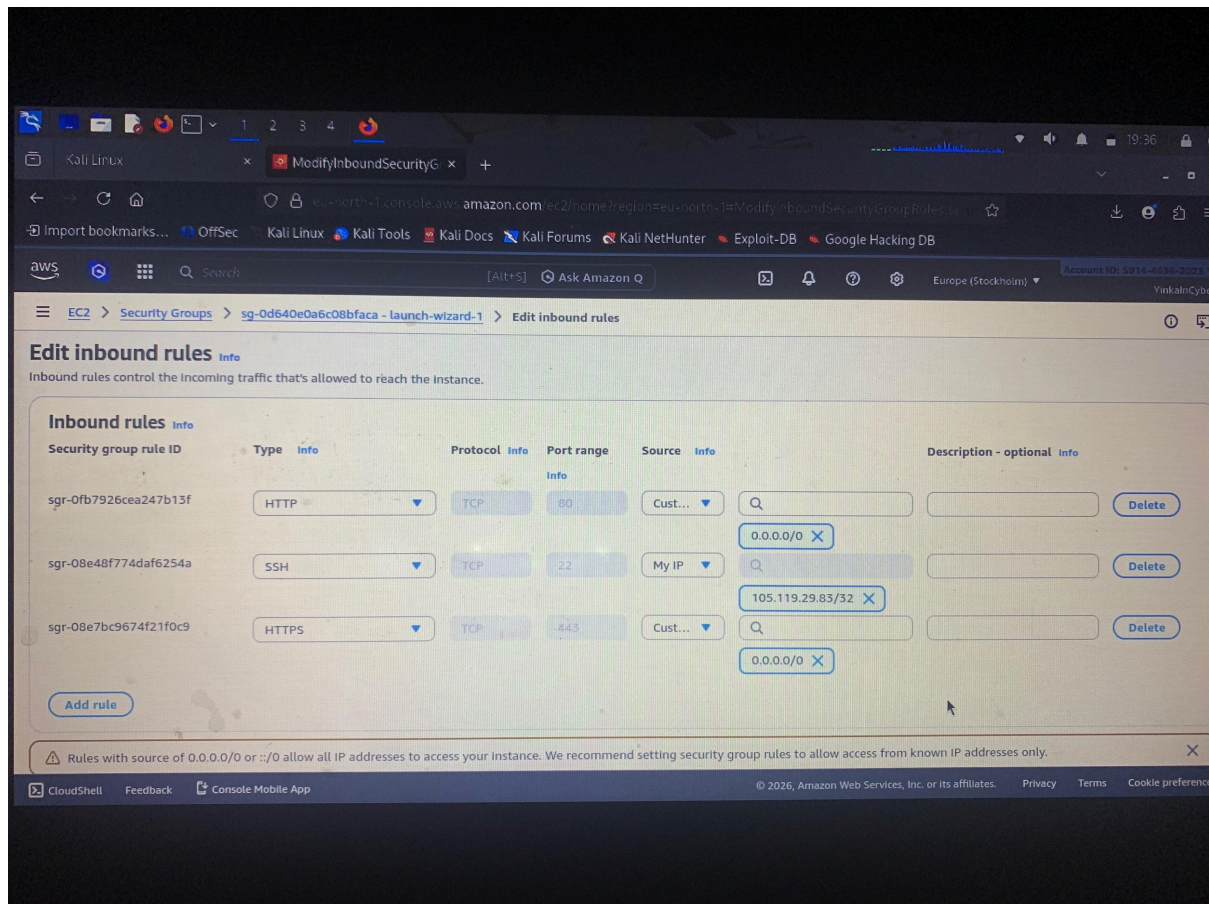
The security group was configured to allow controlled access to the EC2 instance.

Inbound rules include:

- SSH (22): restricted to “My IP” for secure administrative access
- HTTP (80): allowed from 0.0.0.0/0 for public web access
- HTTPS (443): allowed from 0.0.0.0/0 for secure web access

TYPE	PORT	Source
SSH	22	My IP (initial) / 0.0.0.0/0 (temporary troubleshooting)
HTTP	80	0.0.0.0/0

HTTPS	443	0.0.0.0/0
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4. Network Configuration (VPC, Subnet & Routing)

The EC2 instance was launched inside the default AWS Virtual Private Cloud (VPC), which provides networking capabilities for cloud resources.

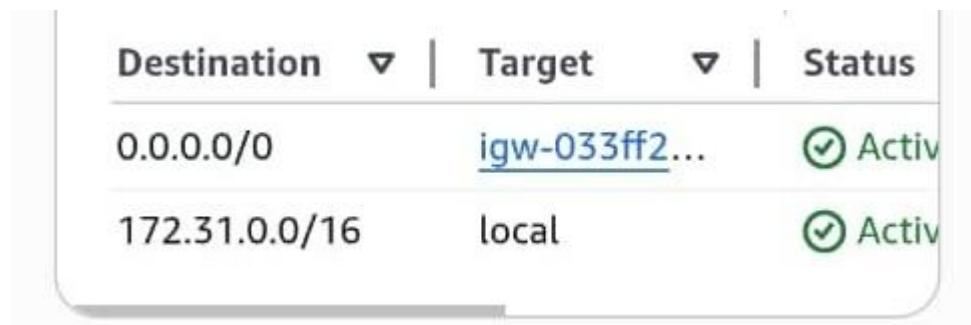
A public subnet was used to ensure the instance could be accessed over the internet. Internet connectivity was enabled through an Internet Gateway attached to the VPC.

A route table was configured and associated with the subnet to allow outbound and inbound internet traffic.

Route Table Configuration:

- Destination: 0.0.0.0/0
- Target: Internet Gateway (IGW)

This configuration allows the EC2 instance to communicate with the internet and be accessible via its public IPv4 address.

A screenshot of the AWS Network Manager console showing a route table configuration. The table has three columns: Destination, Target, and Status. The first row shows Destination 0.0.0.0/0, Target igw-033ff2..., and Status Active. The second row shows Destination 172.31.0.0/16, Target local, and Status Active.

Destination ▼	Target ▼	Status
0.0.0.0/0	igw-033ff2...	✓ Active
172.31.0.0/16	local	✓ Active

5. Server Access (SSH Connection)

The EC2 instance was accessed remotely using SSH from a local machine (Kali Linux environment).
With this command: `ssh -i ~/Downloads/prestashop-key.pem ubuntu@<Public-IP>`

Successful authentication confirmed secure remote access to the server.


```
ubuntu@ip-172-31-6-33:~$ cat ~/Downloads/pretashop-key.pem
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed May 13 07:58:21 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.9% of 6.61GB Processes:      115
Memory usage: 28%          Users logged in: 0
Swap usage:  0%            IPv4 address for ens5: 172.31.6.33

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-6-33:~$
```

6. Apache Web Server Installation

Apache was installed to serve web content on the EC2 instance.

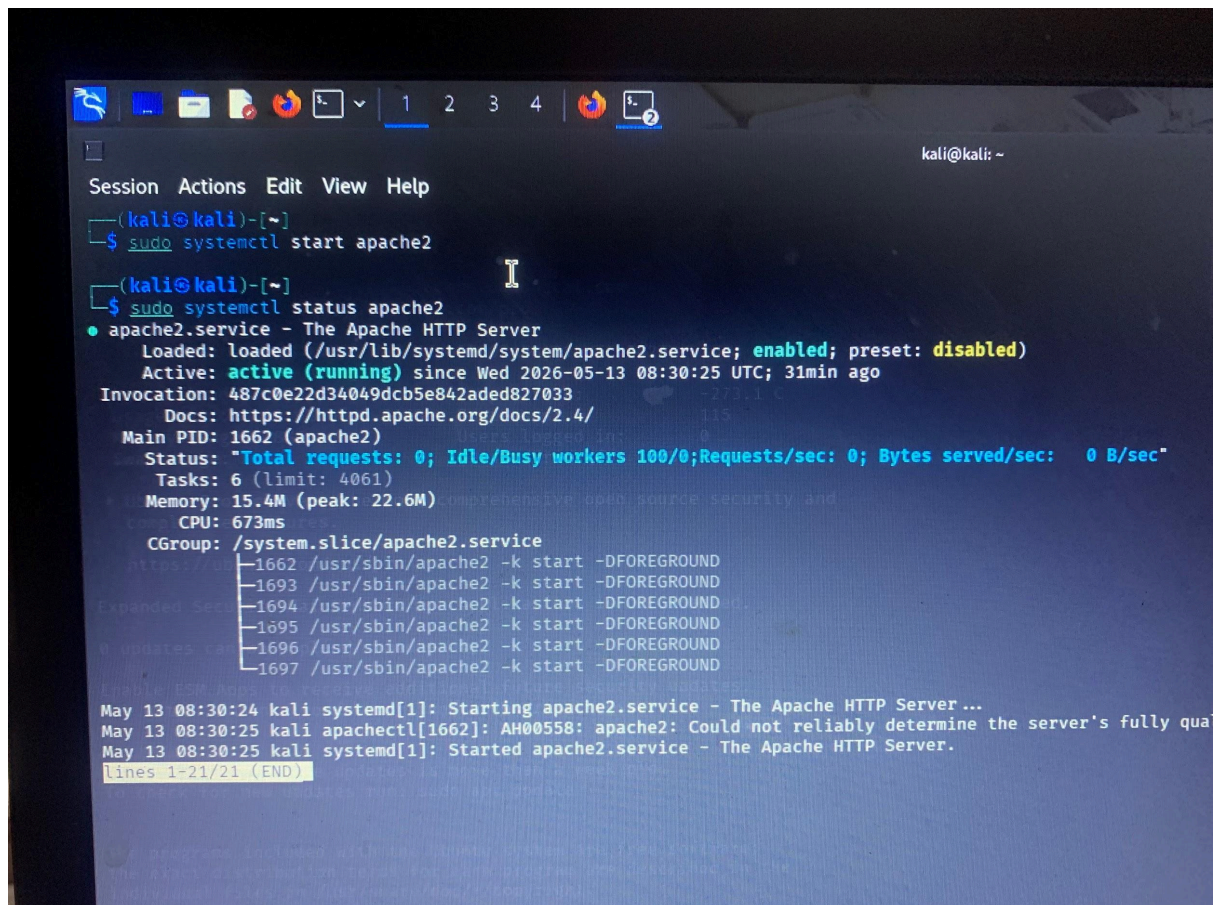
Installation Commands: `sudo apt update`

`sudo apt install apache2 -y`

Service Status Check:

`sudo systemctl status apache2`

Apache was confirmed running successfully.

A terminal window on a Kali Linux system. The user has run 'sudo systemctl start apache2' and 'sudo systemctl status apache2'. The status output shows that the 'apache2.service' is loaded, enabled, and active (running) since May 13, 2026, at 08:30:25 UTC. It lists various metrics like Total requests (0), Idle/Busy workers (100/0), and Memory usage (15.4M). It also shows the CGroup and the list of processes running under the 'apache2' user.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ sudo systemctl start apache2  
(kali@kali)-[~]  
$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)  
   Active: active (running) since Wed 2026-05-13 08:30:25 UTC; 31min ago  
  Invocation: 487c0e22d34049dcb5e842aded827033  
     Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 1662 (apache2)  
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"  
   Tasks: 6 (limit: 4061)  
  Memory: 15.4M (peak: 22.6M)  
     CPU: 673ms  
   CGroup: /system.slice/apache2.service  
           └─1662 /usr/sbin/apache2 -k start -DFOREGROUND  
             └─1693 /usr/sbin/apache2 -k start -DFOREGROUND  
               └─1694 /usr/sbin/apache2 -k start -DFOREGROUND  
                 └─1695 /usr/sbin/apache2 -k start -DFOREGROUND  
                   └─1696 /usr/sbin/apache2 -k start -DFOREGROUND  
                     └─1697 /usr/sbin/apache2 -k start -DFOREGROUND  
May 13 08:30:24 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...  
May 13 08:30:25 kali apachectl[1662]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the documentation at http://httpd.apache.org/docs/2.4/faq.html#fqdn  
May 13 08:30:25 kali systemd[1]: Started apache2.service - The Apache HTTP Server.  
lines 1-21/21 (END)
```

7. Web Server Testing and Accessibility

The Apache web server was tested using the EC2 instance public IPv4 address: <http://13.50.245.17>

Initial internal testing confirmed that the Apache service is running successfully on the EC2 instance using local access (localhost), indicating that the web server is properly installed and active.

However, external browser access from the internet was not fully successful during the testing phase. The following troubleshooting steps were performed:

- Verified EC2 instance status (running state confirmed)
- Confirmed security group configuration allowing HTTP (port 80)
- Confirmed route table includes Internet Gateway (IGW)
- Verified Apache service is active and running on the instance
- Established successful SSH connection to the instance

Despite these configurations, browser access to the public IP was not completed successfully within the testing timeframe.



This site can't be reached

13.50.245.17 refused to connect.

Try:

Checking the connection

ERR_CONNECTION_REFUSED

Reload

Details

8. PrestaShop Deployment (Preparation Stage)

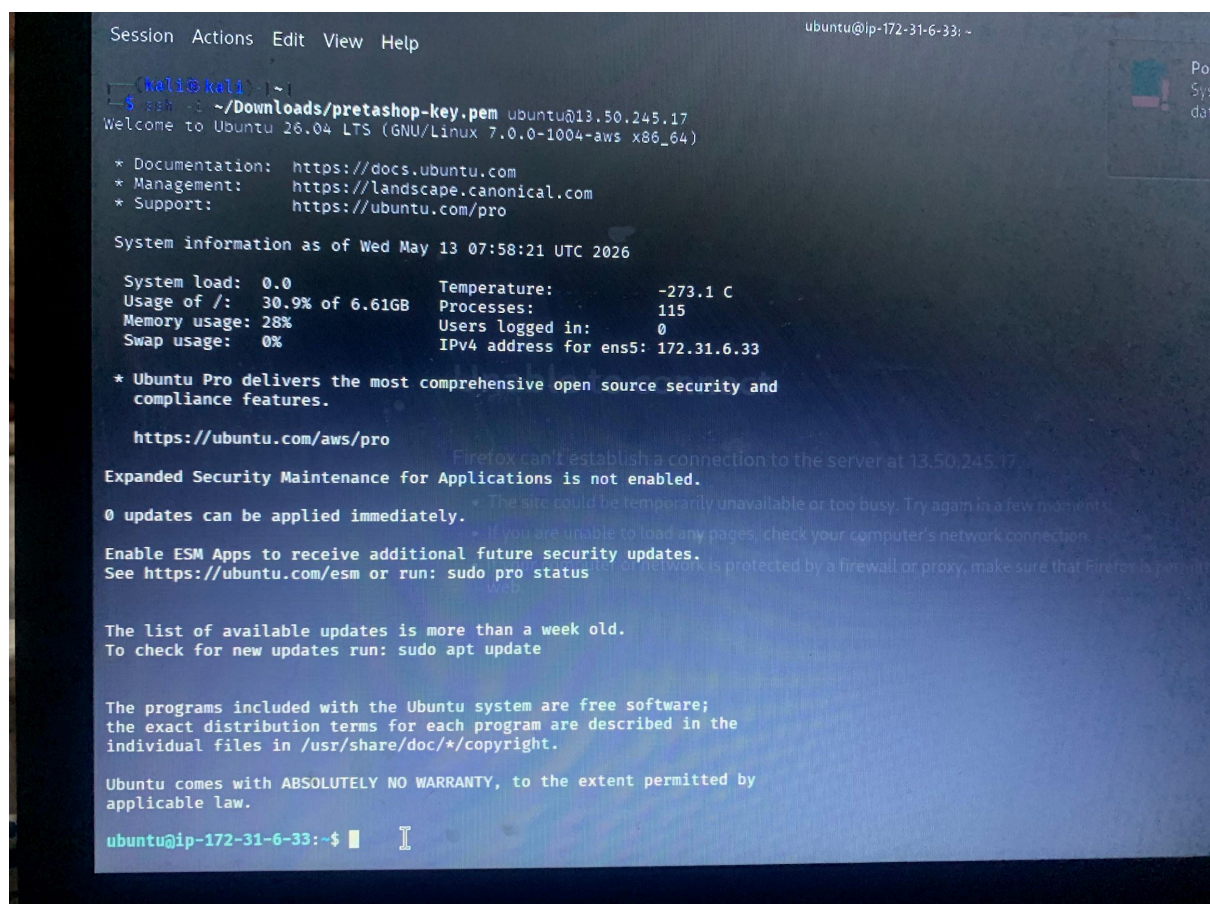
[PrestaShop Official Website](#) was identified as the target application for deployment on the EC2 instance.

The server environment was prepared by successfully installing and configuring Apache on the Ubuntu EC2 instance. This provided the required web server foundation for hosting the application.

However, the full PrestaShop installation was not completed within the scope and timeframe of this assessment.

The following components were identified as the next deployment steps:

- Installation of required PHP dependencies
- Configuration of a separate database using AWS RDS (MySQL)
- Completion of the PrestaShop web-based installation wizard



```
Session Actions Edit View Help
ubuntu@ip-172-31-6-33: ~

kali@kali: ~$ ssh -i ~/Downloads/prestashop-key.pem ubuntu@13.50.245.17
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
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   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
  - The site could be temporarily unavailable or too busy. Try again in a few moments.
  - If you are unable to load any pages, check your computer's network connection.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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ubuntu@ip-172-31-6-33:~$
```

9. Database Architecture Requirement

In line with cloud architecture best practices, the database layer is designed to be hosted separately from the application server.

For this deployment, AWS RDS (MySQL) was identified as the preferred database solution instead of installing the database directly on the EC2 instance.

This approach follows a multi-tier architecture model consisting of:

- Presentation Layer: Web browser access
- Application Layer: EC2 Ubuntu server running Apache
- Data Layer: AWS RDS (MySQL database service)

Benefits of this architecture include:

- Security: Separation of database from public-facing server reduces exposure risk
- Scalability: Database resources can be scaled independently of the application server
- Reliability: Managed database service improves uptime and reduces maintenance overhead
- Best Practice Compliance: Aligns with standard cloud deployment architecture principles

Note: No RDS instance was deployed during this phase; this section describes the intended architecture design.

11. Explanation of External Access Failure

During testing, external browser access to the Apache web server using the EC2 public IP address was not successful. Several potential causes were identified through troubleshooting analysis.

Possible Causes:

- Apache Binding to Localhost:
Apache may be configured to listen only on 127.0.0.1 instead of all interfaces (0.0.0.0), preventing external access.
- Security Group Misconfiguration:
Although HTTP port 80 was opened, incorrect or temporary firewall settings could restrict inbound traffic.

- Network ACL Restrictions:

AWS Network Access Control Lists (ACLs) may block inbound or outbound traffic if not properly configured.

- Missing or Changing Public IP (Elastic IP not used):

The instance uses an auto-assigned public IPv4 address, which may change after restart, affecting accessibility.

- UFW (Linux Firewall) Configuration:

Ubuntu firewall settings may block incoming HTTP traffic if enabled.

- Browser or DNS Caching Issues:

Local browser caching or ISP-level DNS resolution delays may affect connectivity testing.

12. Cybersecurity Hardening Measures

To improve the security posture of the deployed EC2 environment, several best practices were considered:

- SSH Key Authentication:

Secure login was implemented using SSH key pairs instead of password authentication.

- Disabling Root Login (Recommended):

Direct root login should be disabled to reduce attack surface.

- System Updates:

Regular updates should be applied using: `sudo apt update && sudo apt upgrade -y`

13. Conclusion

This project demonstrates the deployment of a cloud-based web server using Amazon Web Services (AWS). An Ubuntu EC2 instance was successfully created and configured to host a web server environment.

Key implementations in this project include:

- Creation and configuration of an AWS EC2 instance
- Setup of security groups to manage inbound traffic

- Configuration of VPC routing for internet connectivity
- Installation and activation of the Apache web server
- Verification of internal server functionality through SSH access

Although external browser access to the web server was not fully successful during testing, the core infrastructure setup and server configuration were completed and validated.

The project also highlights best practices in cloud architecture, including separation of application and database layers using AWS RDS and implementation of basic cybersecurity hardening principles.

Overall, this work demonstrates practical understanding of cloud computing, Linux server management, and AWS infrastructure deployment.

References

- Amazon Web Services Documentation – EC2
<https://docs.aws.amazon.com/ec2/>
- Amazon Web Services Documentation – VPC
<https://docs.aws.amazon.com/vpc/>
- Apache HTTP Server Documentation
<https://httpd.apache.org/docs/>
- PrestaShop Official Website
<https://www.prestashop.com/en>